

PHY 321 Midterm 2

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```
In [59]: import numpy as np
from math import *
import matplotlib.pyplot as plt
import pandas as pd
%matplotlib inline
plt.style.use('seaborn-v0_8-colorblind')
```

Undamped and undriven case (20 points)

- 1a. (5 pt) Consider the undriven and undamped case, $\delta = \gamma = 0$. What potential would give rise to this kind of equation of motion? Demonstrate that potential gives the expected equation of motion.

Answer to 1a. is attached

- 1b. (5 pt) Find the equilibrium points of the system for the undriven and undamped case. What are the conditions for stable and unstable equilibrium points? Plot the potential for some choices of α and β to illustrate your findings.

Photos for conditions on stable and unstable equilibrium points are attached

In order for $x = (-\alpha/\beta)^{1/2}$ to exist, either α or β has to be negative and the other positive.

If $\alpha > 0$ then $x=0$ is a stable equilibrium point and $x = \pm (-\alpha/\beta)^{1/2}$ is an unstable equilibrium point.

If $\alpha < 0$ then $x=0$ is an unstable equilibrium point and $x = \pm (-\alpha/\beta)^{1/2}$ is a stable equilibrium point.

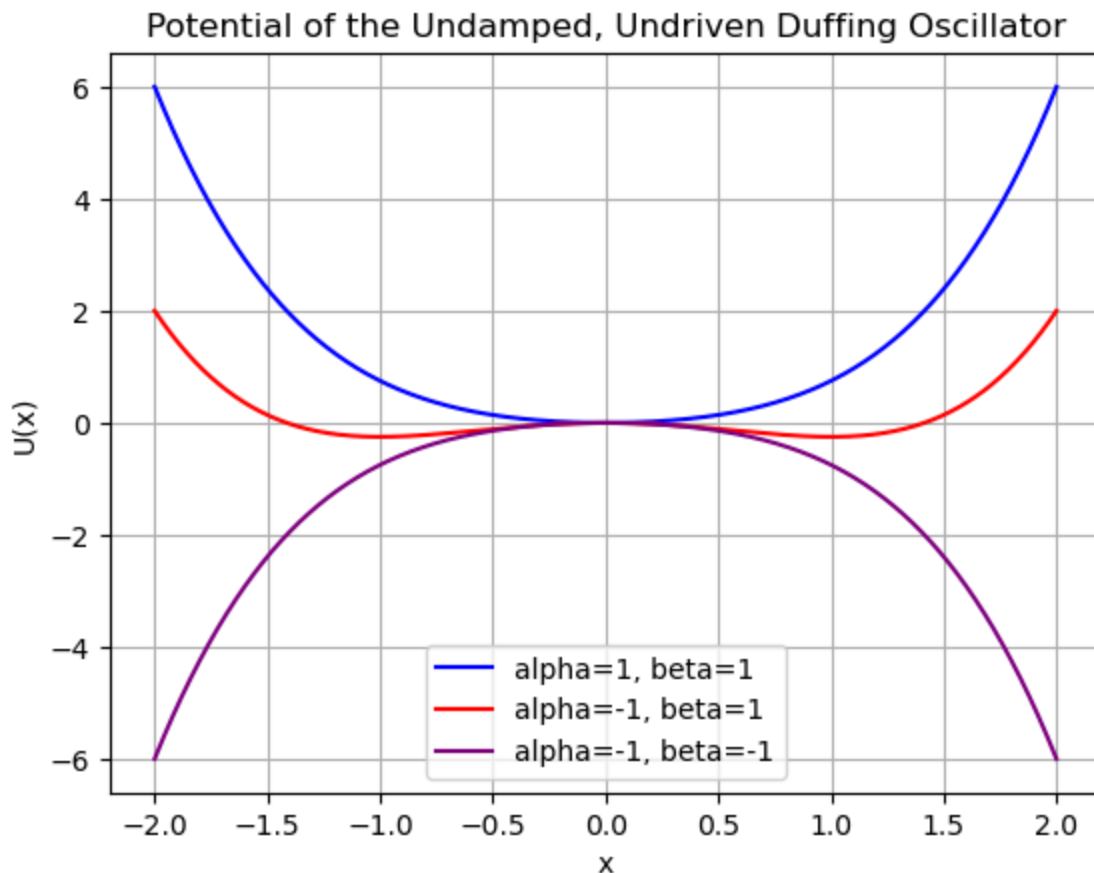
```
In [60]: alpha1 = 1
beta1 = 1
alpha2 = -1
beta2 = -1
m = 1
x = np.linspace(-2,2,1000)
U1 = m * alpha1 * x**2/2 + m * beta1 * x**4/4
```

```

U2 = m * alpha2 * x**2/2 + m * beta1 * x**4/4
U3 = m * alpha2 * x**2/2 + m * beta2 * x**4/4

plt.plot(x, U1, color='b', label='alpha=1, beta=1')
plt.plot(x, U2, color='r', label='alpha=-1, beta=1')
plt.plot(x, U3, color='purple', label='alpha=-1, beta=-1')
plt.xlabel('x')
plt.ylabel('U(x)')
plt.title('Potential of the Undamped, Undriven Duffing Oscillator')
plt.grid()
plt.legend()
plt.show()

```



For $\alpha = 1, \beta = 1$, we see a stable equilibrium point at $x=0$ since it lies at the bottom of the potential well. There are no other equilibrium points in this case since both α and β are positive, the other equilibrium points are imaginary. This agrees with the conditions derived.

For $\alpha = -1, \beta = 1$, we see a unstable equilibrium point at $x=0$ since it lies at the top of the local potential well. There are two other stable equilibrium points ($x = -1, 1$) corresponding to $x = \pm(-\alpha/\beta)^{(1/2)}$ since α is positive and β is negative.

For $\alpha = -1, \beta = -1$, we see $x=0$ is an unstable equilibrium point since it's at the top of the local potential well. No other equilibrium points exist agreeing with the

conditions.

- 1c. (5 pt) Plot the phase space for the undriven and undamped case. What are the trajectories of the system? What is the behavior of the system? Does it match your expectations from the potential? *The figure above only starts this discussion.*

```
In [61]: def DO_undamped_undriven(X,V):

    dX = V
    dV = -alpha1 * X - beta1 * X**3

    return dX, dV

def generate_phase_space(x_lim, v_lim, grid_size):

    x = np.linspace(x_lim[0], x_lim[1], grid_size)
    v = np.linspace(v_lim[0], v_lim[1], grid_size)

    X, V = np.meshgrid(x, v)

    dX, dV = DO_undamped_undriven(X, V)

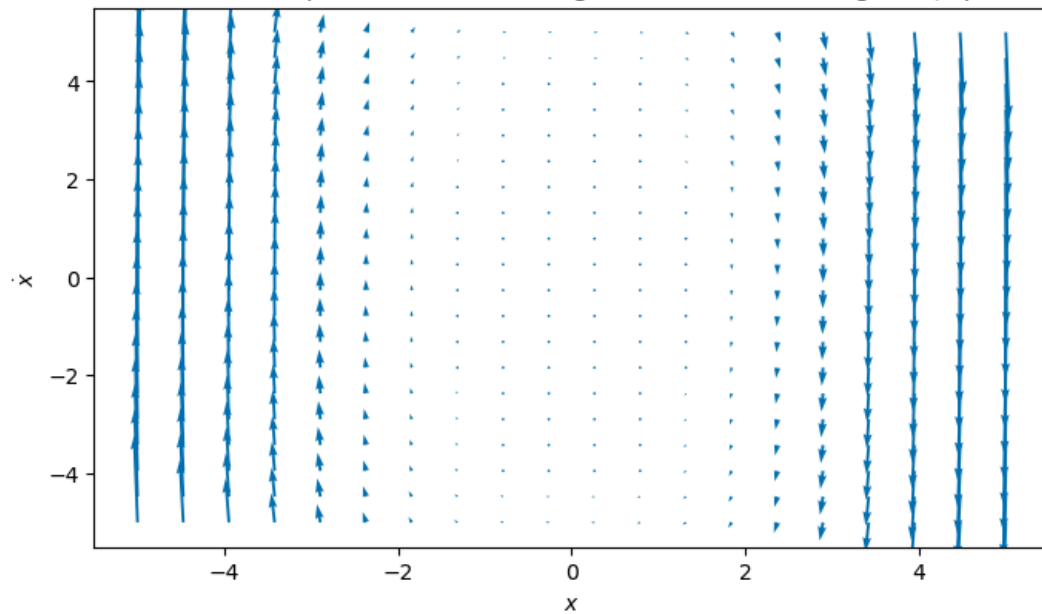
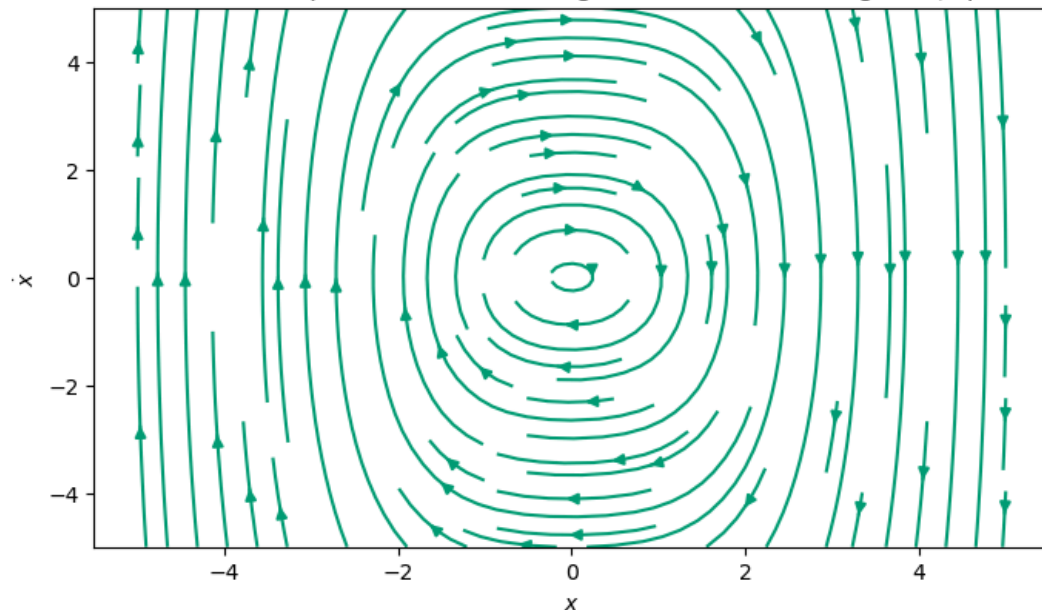
    return X, V, dX, dV

x_lim = (-5, 5)
v_lim = (-5, 5)
grid_size = 20
X, V, dX, dV = generate_phase_space(x_lim, v_lim, grid_size)
```

```
In [62]: fig, axs = plt.subplots(2, 1, figsize=(8, 10)) # Two rows, one column

axs[0].quiver(X, V, dX, dV, color='C0')
axs[0].set_title('Quiver Plot of the Undamped, Undriven Duffing Oscillator Phase Di
axs[0].set_xlabel(r'$x$')
axs[0].set_ylabel(r'$\dot{x}$')

axs[1].streamplot(X, V, dX, dV, color='C1')
axs[1].set_title('Stream Plot of the Undamped, Undriven Duffing Oscillator Phase Di
axs[1].set_xlabel(r'$x$')
axs[1].set_ylabel(r'$\dot{x}$');
plt.show()
```

Quiver Plot of the Undamped, Undriven Duffing Oscillator Phase Diagram ($\alpha=1, \beta=1$)Stream Plot of the Undamped, Undriven Duffing Oscillator Phase Diagram ($\alpha=1, \beta=1$)

The phase diagram for an undamped, undriven duffing oscillator with $\alpha = 1, \beta = 1$ shows oscillations around $x = 0$ similar to the similar harmonic oscillator. An initial position and velocity of a system would oscillate and reach the same point. I expected this since the graph of the potential with these parameters resembled the SHO potential and they both only have a stable equilibrium point of $x = 0$.

```
In [63]: def DO_undamped_undriven2(X,V):

    dX = V
    dV = -alpha2 * X - beta1 * X**3

    return dX, dV

def generate_phase_space2(x_lim, v_lim, grid_size):
```

```

x = np.linspace(x_lim[0], x_lim[1], grid_size)
v = np.linspace(v_lim[0], v_lim[1], grid_size)

X2, V2 = np.meshgrid(x, v)

dX2, dV2 = D0_undamped_undriven2(X2, V2)

return X2, V2, dX2, dV2

x_lim = (-1.5, 1.5)
v_lim = (-5, 5)
grid_size = 20
X2, V2, dX2, dV2 = generate_phase_space2(x_lim, v_lim, grid_size)

```

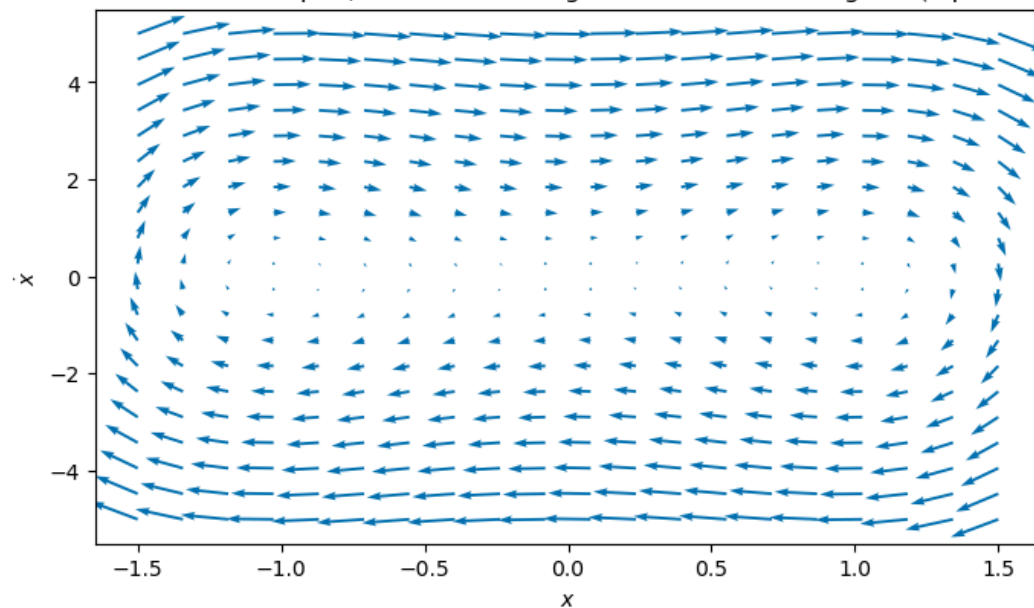
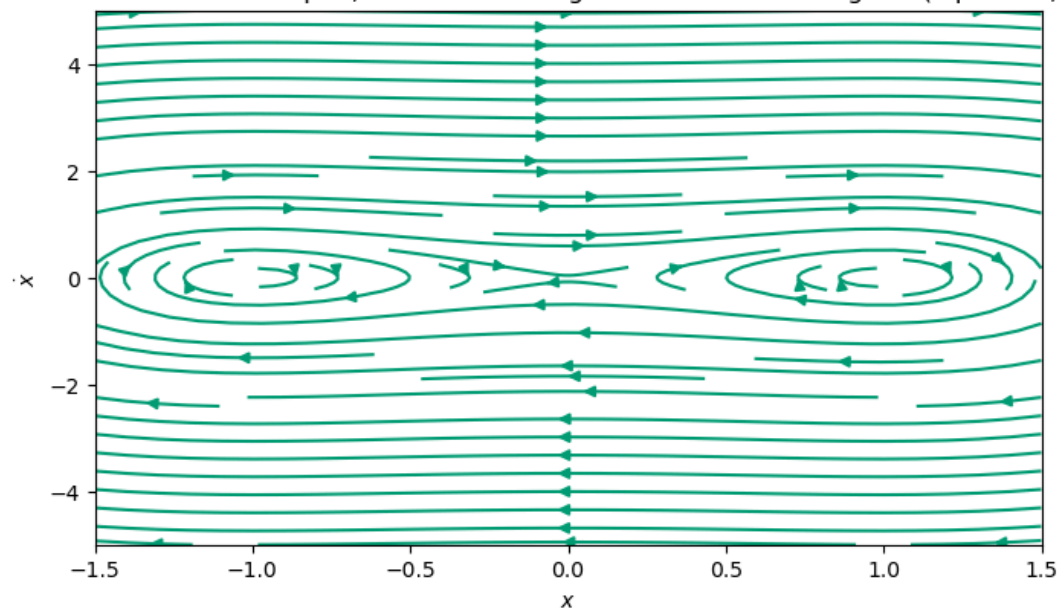
```

In [64]: fig, axs = plt.subplots(2, 1, figsize=(8, 10)) # Two rows, one column

axs[0].quiver(X2, V2, dX2, dV2, color='C0')
axs[0].set_title('Quiver Plot of the Undamped, Undriven Duffing Oscillator Phase Di
axs[0].set_xlabel(r'$x$')
axs[0].set_ylabel(r'$\dot{x}$')

axs[1].streamplot(X2, V2, dX2, dV2, color='C1')
axs[1].set_title('Stream Plot of the Undamped, Undriven Duffing Oscillator Phase Di
axs[1].set_xlabel(r'$x$')
axs[1].set_ylabel(r'$\dot{x}$');
plt.show()

```

Quiver Plot of the Undamped, Undriven Duffing Oscillator Phase Diagram ($\alpha=-1, \beta=1$)Stream Plot of the Undamped, Undriven Duffing Oscillator Phase Diagram ($\alpha=-1, \beta=1$)

The phase diagram for the case $\alpha = -1, \beta = 1$ shows trajectories that oscillate around the equilibrium points -1 and 1 . A system with an initial position and velocity near those points oscillate and return back to the same position. We also see trajectories passing through or away from $x=0$. This is expected since $x=0$ is an unstable equilibrium point and $x=-1, 1$ are stable equilibrium points from the potential.

```
In [65]: def DO_undamped_undriven3(X,V):
          dX = V
          dV = -alpha2 * X - beta2 * X**3
          return dX, dV

          def generate_phase_space3(x_lim, v_lim, grid_size):
```

```

x = np.linspace(x_lim[0], x_lim[1], grid_size)
v = np.linspace(v_lim[0], v_lim[1], grid_size)

X3, V3 = np.meshgrid(x, v)

dX3, dV3 = D0_undamped_undriven3(X3, V3)

return X3, V3, dX3, dV3

x_lim = (-5, 5)
v_lim = (-5, 5)
grid_size = 20
X3, V3, dX3, dV3 = generate_phase_space3(x_lim, v_lim, grid_size)

```

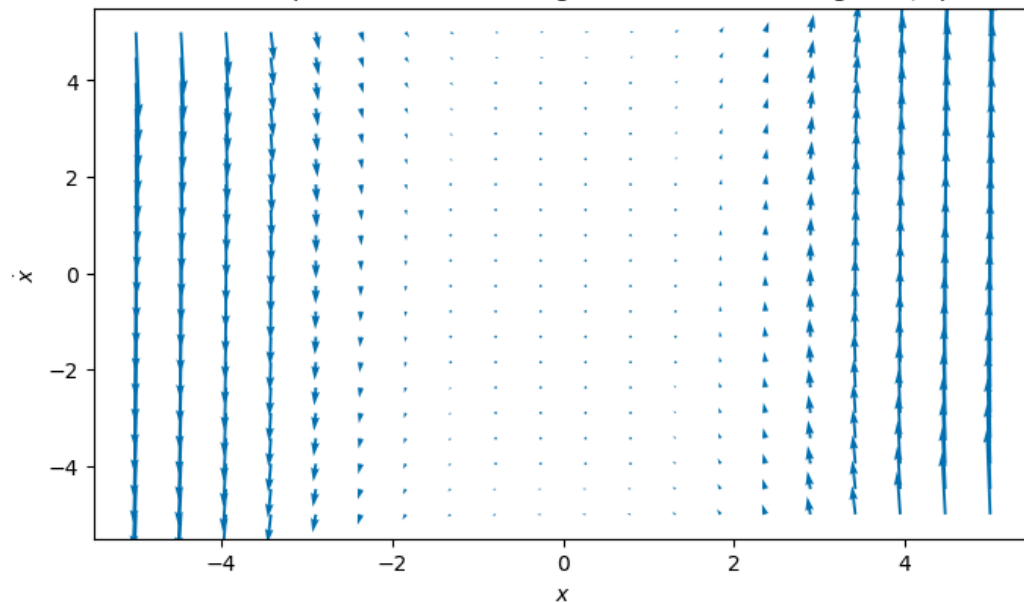
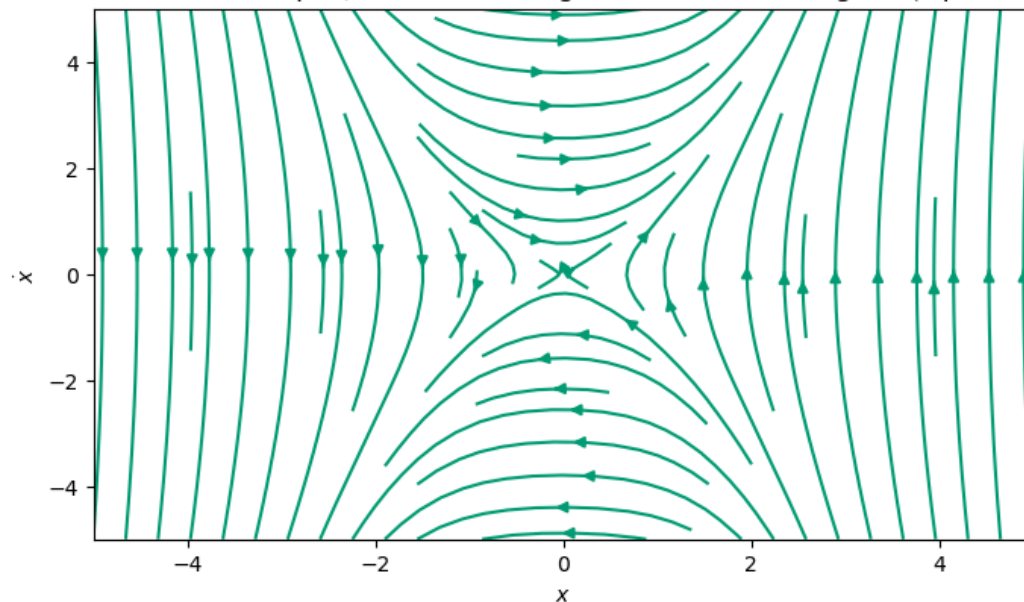
```

In [66]: fig, axs = plt.subplots(2, 1, figsize=(8, 10)) # Two rows, one column

axs[0].quiver(X3, V3, dX3, dV3, color='C0')
axs[0].set_title('Quiver Plot of the Undamped, Undriven Duffing Oscillator Phase Di
axs[0].set_xlabel(r'$x$')
axs[0].set_ylabel(r'$\dot{x}$')

axs[1].streamplot(X3, V3, dX3, dV3, color='C1')
axs[1].set_title('Stream Plot of the Undamped, Undriven Duffing Oscillator Phase Di
axs[1].set_xlabel(r'$x$')
axs[1].set_ylabel(r'$\dot{x}$');
plt.show()

```

Quiver Plot of the Undamped, Undriven Duffing Oscillator Phase Diagram ($\alpha=-1, \beta=-1$)Stream Plot of the Undamped, Undriven Duffing Oscillator Phase Diagram ($\alpha=-1, \beta=-1$)

The phase diagram for the $\alpha = -1, \beta = -1$ case shows all trajectories moving away from $x=0$ with no oscillation. This is expected since $x=0$ is an unstable equilibrium point from the potential.

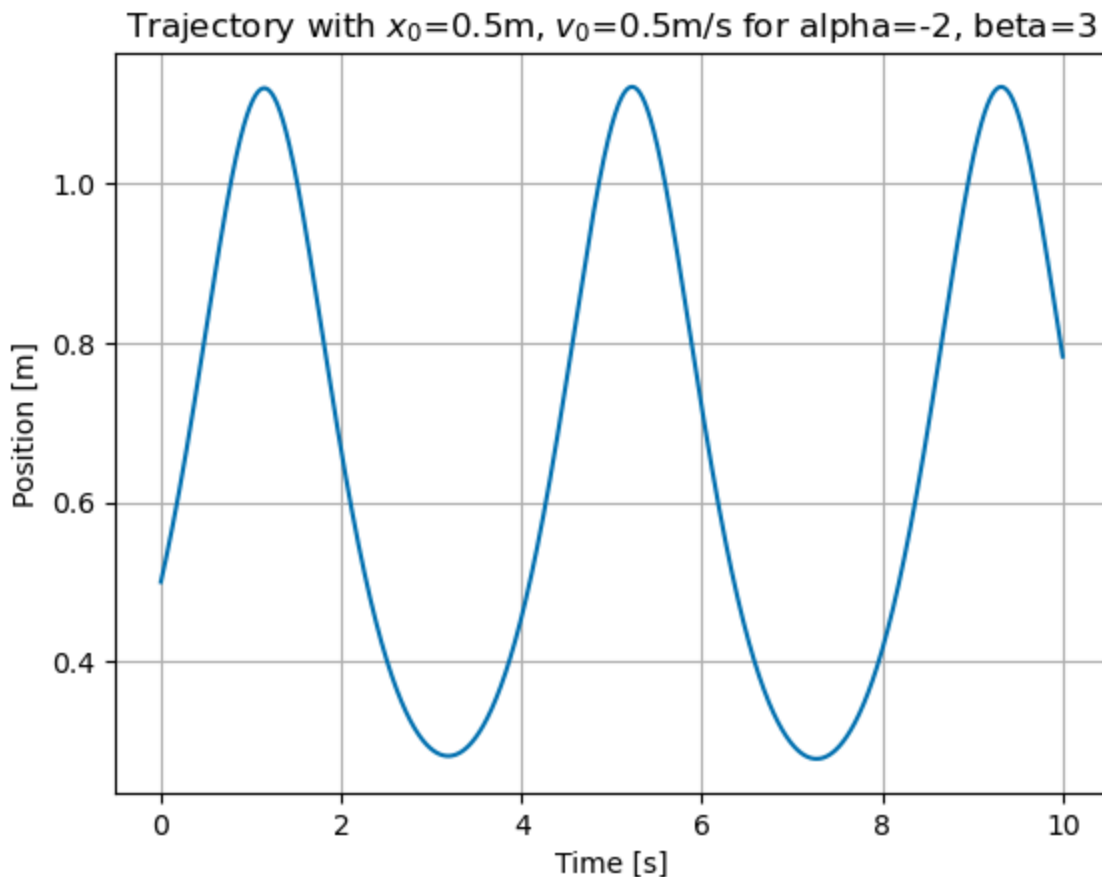
- 1d. (5 pt) **Here you are likely to need a numerical solver.** Solve the equation of motion for the undriven and undamped case for a reasonable choice of α and β and initial conditions. Plot the position as a function of time. What is the behavior of the system? Does it match your expectations from the potential and the phase portrait?

```
In [67]: from scipy.integrate import solve_ivp
```

```
alpha5 = -2
beta5 = 3
```

```
def DO_ud_ud(t, y, alpha5, beta5): #ud_ud represents undamped, undriven
    x, v = y
    return [v, -alpha5 * x - beta5 * x**3]

y0 = [0.5, 0.5] #Initial conditions are chosen to be x_0=0.5m, v_0=0.5m/s
t_span = np.linspace(0, 10, 10000)
sol_ud_ud = solve_ivp(DO_ud_ud, (0,10), y0, args=(alpha5, beta5), t_eval=t_span)
x1 = sol_ud_ud.y[0]
v1 = sol_ud_ud.y[1]
plt.plot(sol_ud_ud.t, x1)
plt.xlabel('Time [s]')
plt.ylabel('Position [m]')
plt.title('Trajectory with $x_0=0.5m$, $v_0=0.5m/s$ for alpha=-2, beta=3')
plt.grid()
plt.show()
```



The particle starts at 0.5m with velocity 0.5m/s and we observe an oscillation around the point $(2/3)^{1/2} = 0.816m$ when calculated using the formula of equilibrium points of the system. This is expected since this point must be a stable equilibrium point when observing the graph of a similar potential using the similar parameters as well as the phase diagram.

Damped and undriven case (15 points)

- 1e. (5 pt) Now consider the damped and undriven case, $\gamma = 0$. What is the equation of motion in this case? Can you develop this equation of motion fully from a potential (like

we did above)? If so, what is the potential? If not, why not?

$$\ddot{x} = -\alpha x - \beta x^3 - \delta \dot{x}$$

This equation of motion cannot be obtained from a potential. The force function is dependent on velocity not dependent on coordinates alone. Thus, the curl of the force is non-zero and the force is not conservative.

- 1f. (5 pt) Plot the phase space for the damped and undriven case. What kinds of trajectories are there?

```
In [68]: alpha6 = 2
beta6 = -4
delta6 = 2

def DO_damped_undriven1(X,V):

    dX = V
    dV = -alpha6 * X - beta6 * X**3 - delta6 * V

    return dX, dV

def generate_phase_space_d_ud1(x_lim, v_lim, grid_size):    #d_ud stands for damped,

    x = np.linspace(x_lim[0], x_lim[1], grid_size)
    v = np.linspace(v_lim[0], v_lim[1], grid_size)

    X6, V6 = np.meshgrid(x, v)

    dX6, dV6 = DO_damped_undriven1(X6, V6)

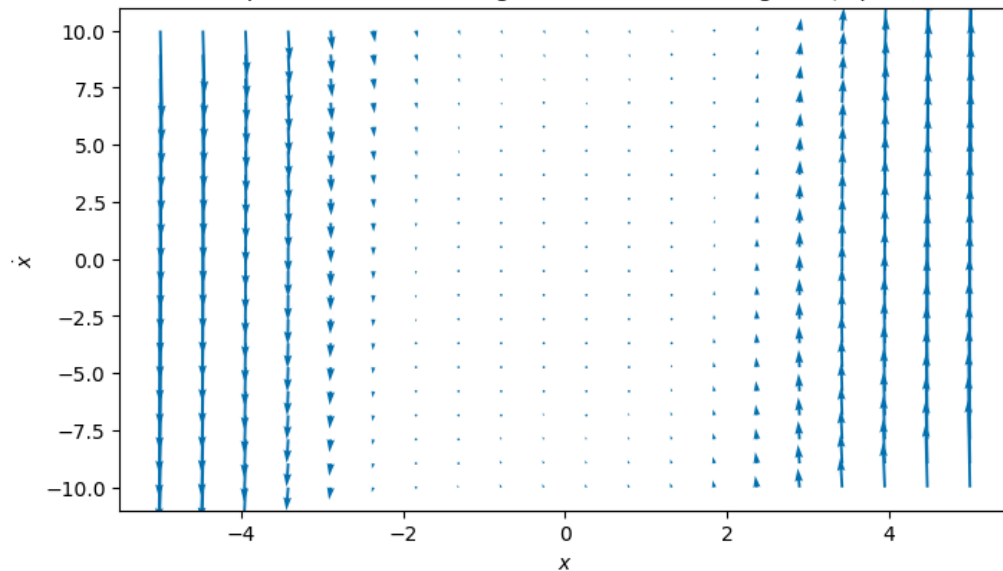
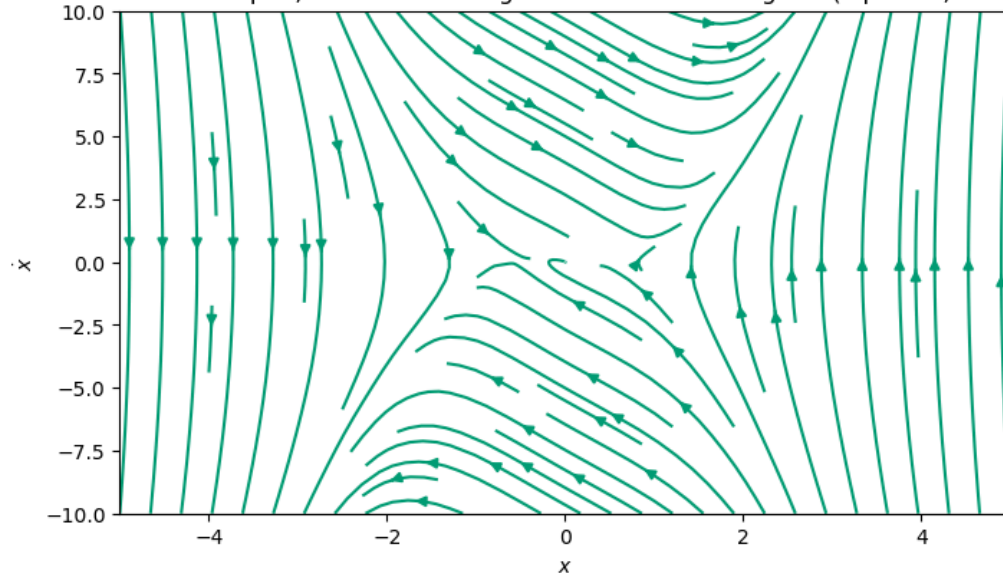
    return X6, V6, dX6, dV6

x_lim = (-5, 5)
v_lim = (-10, 10)
grid_size = 20
X6, V6, dX6, dV6 = generate_phase_space_d_ud1(x_lim, v_lim, grid_size)
```

```
In [69]: fig, axs = plt.subplots(2, 1, figsize=(8, 10))    # Two rows, one column

axs[0].quiver(X6, V6, dX6, dV6, color='C0')
axs[0].set_title('Quiver Plot of the Damped, Undriven Duffing Oscillator Phase Diag
axs[0].set_xlabel(r'$x$')
axs[0].set_ylabel(r'$\dot{x}$')

axs[1].streamplot(X6, V6, dX6, dV6, color='C1')
axs[1].set_title('Stream Plot of the Damped, Undriven Duffing Oscillator Phase Diag
axs[1].set_xlabel(r'$x$')
axs[1].set_ylabel(r'$\dot{x}$');
plt.show()
```

Quiver Plot of the Damped, Undriven Duffing Oscillator Phase Diagram ($\alpha=2, \beta=-4, \delta=2$)Stream Plot of the Damped, Undriven Duffing Oscillator Phase Diagram ($\alpha=2, \beta=-4, \delta=2$)

In the case where $\alpha=2, \beta=-4, \delta=2$, the trajectories move away from $x=0$ showing that $x=0$ is an unstable equilibrium point.

```
In [70]: beta7 = 10

def DO_damped_undriven2(X,V):

    dX = V
    dV = -alpha6 * X - beta7 * X**3 - delta6 * V

    return dX, dV

def generate_phase_space_d_ud2(x_lim, v_lim, grid_size):    #d_ud stands for damped,

    x = np.linspace(x_lim[0], x_lim[1], grid_size)
    v = np.linspace(v_lim[0], v_lim[1], grid_size)

    X7, V7 = np.meshgrid(x, v)
```

```

dX7, dV7 = D0_damped_undriven2(X7, V7)

return X7, V7, dX7, dV7

x_lim = (-5, 5)
v_lim = (-10, 10)
grid_size = 20
X7, V7, dX7, dV7 = generate_phase_space_d_ud2(x_lim, v_lim, grid_size)

```

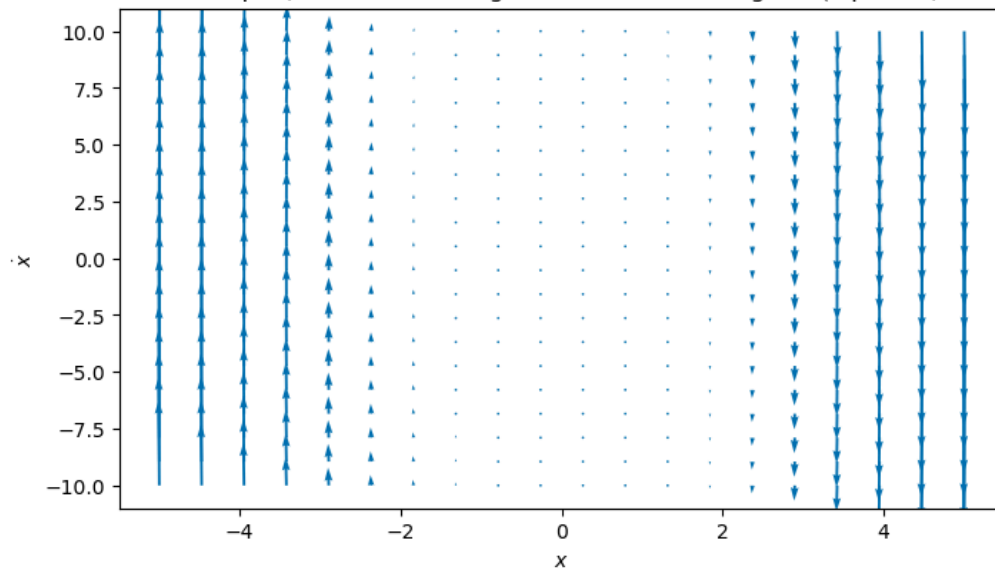
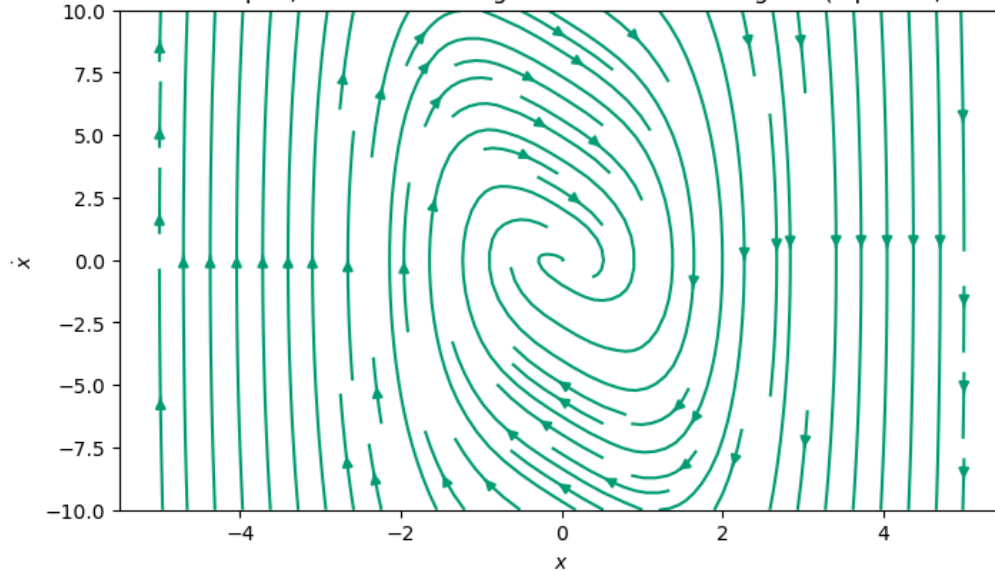
```

In [71]: fig, axs = plt.subplots(2, 1, figsize=(8, 10)) # Two rows, one column

axs[0].quiver(X7, V7, dX7, dV7, color='C0')
axs[0].set_title('Quiver Plot of the Damped, Undriven Duffing Oscillator Phase Diag
axs[0].set_xlabel(r'$x$')
axs[0].set_ylabel(r'$\dot{x}$')

axs[1].streamplot(X7, V7, dX7, dV7, color='C1')
axs[1].set_title('Stream Plot of the Damped, Undriven Duffing Oscillator Phase Diag
axs[1].set_xlabel(r'$x$')
axs[1].set_ylabel(r'$\dot{x}$');
plt.show()

```

Quiver Plot of the Damped, Undriven Duffing Oscillator Phase Diagram ($\alpha=2, \beta=10, \delta=2$)Stream Plot of the Damped, Undriven Duffing Oscillator Phase Diagram ($\alpha=2, \beta=10, \delta=2$)

In the case where β is positive, we see spiral trajectories towards $x = 0$ suggesting that $x = 0$ is a stable equilibrium point.

```
In [72]: alpha7 = 20

def DO_damped_undriven3(X,V):

    dX = V
    dV = -alpha7 * X - beta6 * X**3 - delta6 * V

    return dX, dV

def generate_phase_space_d_ud3(x_lim, v_lim, grid_size): #d_ud stands for damped,

    x = np.linspace(x_lim[0], x_lim[1], grid_size)
    v = np.linspace(v_lim[0], v_lim[1], grid_size)

    X8, V8 = np.meshgrid(x, v)
```

```

    dX8, dV8 = D0_damped_undriven3(X8, V8)

    return X8, V8, dX8, dV8

x_lim = (-5, 5)
v_lim = (-10, 10)
grid_size = 20
X8, V8, dX8, dV8 = generate_phase_space_d_ud3(x_lim, v_lim, grid_size)

```

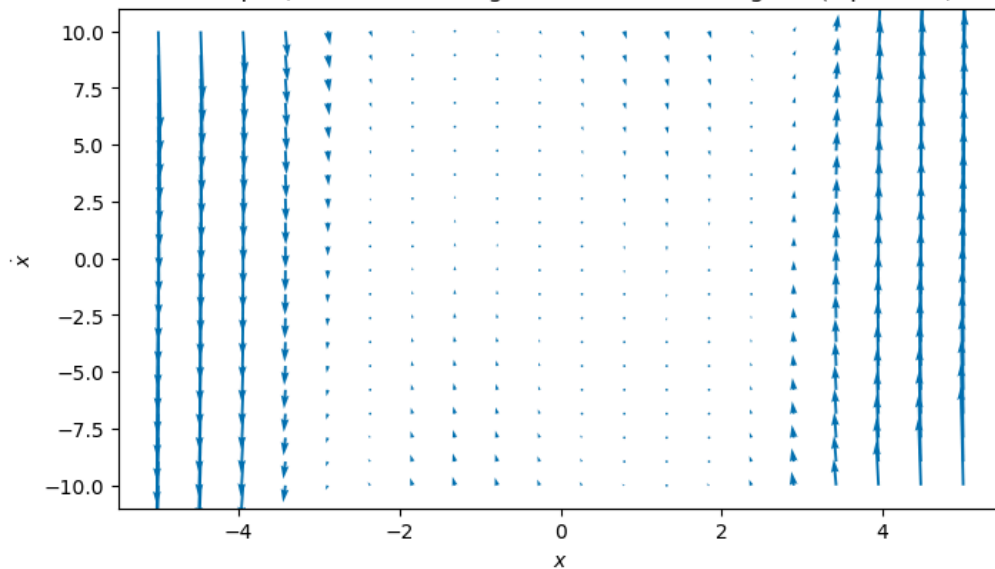
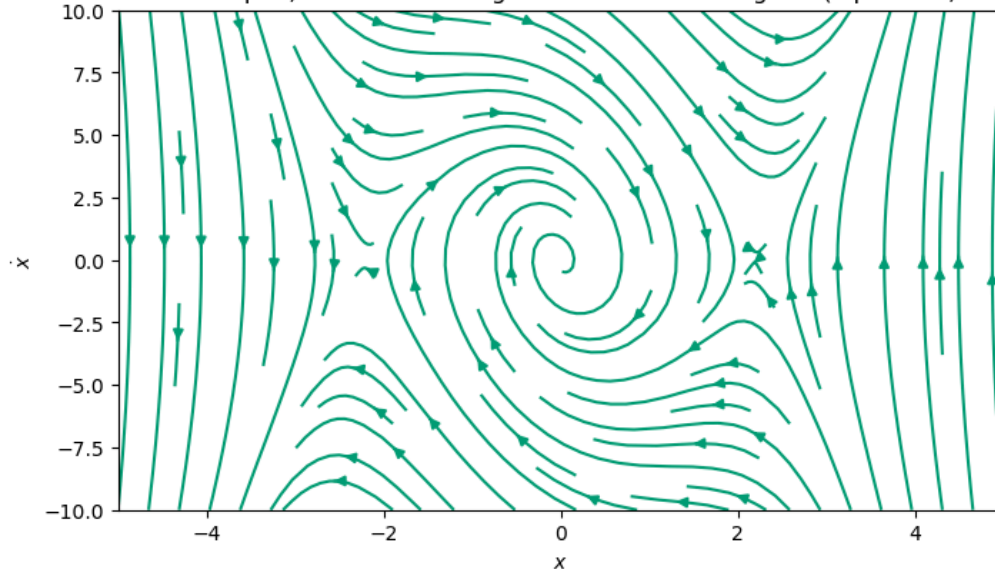
```

In [73]: fig, axs = plt.subplots(2, 1, figsize=(8, 10)) # Two rows, one column

axs[0].quiver(X8, V8, dX8, dV8, color='C0')
axs[0].set_title('Quiver Plot of the Damped, Undriven Duffing Oscillator Phase Diag
axs[0].set_xlabel(r'$x$')
axs[0].set_ylabel(r'$\dot{x}$')

axs[1].streamplot(X8, V8, dX8, dV8, color='C1')
axs[1].set_title('Stream Plot of the Damped, Undriven Duffing Oscillator Phase Diag
axs[1].set_xlabel(r'$x$')
axs[1].set_ylabel(r'$\dot{x}$');
plt.show()

```

Quiver Plot of the Damped, Undriven Duffing Oscillator Phase Diagram ($\alpha=20, \beta=4, \delta=2$)Stream Plot of the Damped, Undriven Duffing Oscillator Phase Diagram ($\alpha=20, \beta=4, \delta=2$)

When α is positive, we see spirals towards $x=0$ showing $x=0$ is a stable equilibrium point. There are trajectories moving away from two points which imply they are unstable equilibrium points. This is similar to a damped pendulum since α dominates over other constants.

- 1g. (5 pt) **Here you are likely to need a numerical solver.** Solve the equation of motion for the damped and undriven case for a reasonable choice of α , β , δ , and initial conditions. Plot the position as a function of time. What is the behavior of the system? Does it match your expectations from the phase portrait?

```
In [74]: alpha9 = 3
         beta9 = 4
         delta9 = 2

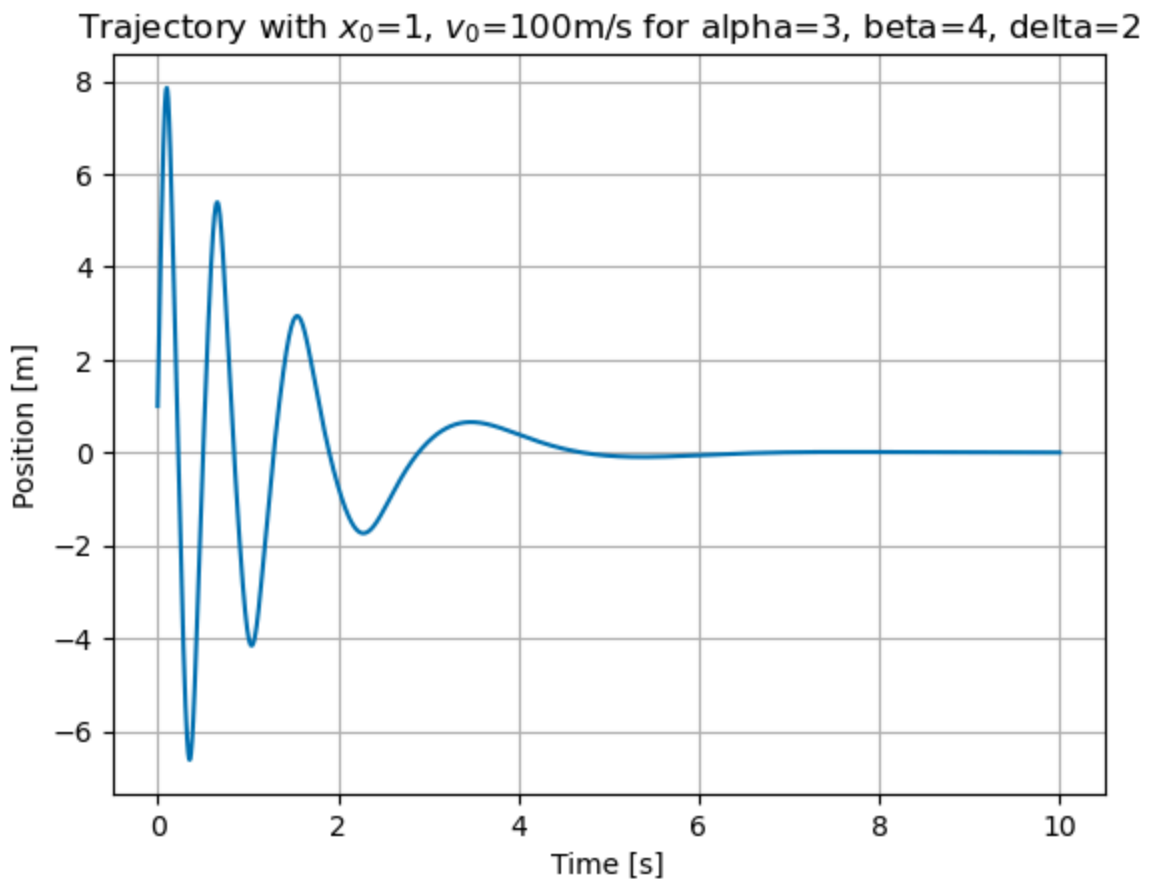
         def DO_d_ud(t, y, alpha5, beta5, delta9): #d_ud represents damped, undriven
             x, v = y
```

```

return [v, -alpha9 * x - beta9 * x**3 - delta9 * v]

y0 = [1, 100] #Initial conditions are chosen to be x_0=1m, v_0=100m/s
t_span = np.linspace(0, 10, 10000)
sol_d_ud = solve_ivp(D0_d_ud, (0,10), y0, args=(alpha9, beta9, delta9), t_eval=t_sp
x2 = sol_d_ud.y[0]
v2 = sol_d_ud.y[1]
plt.plot(sol_d_ud.t, x2)
plt.xlabel('Time [s]')
plt.ylabel('Position [m]')
plt.title('Trajectory with $x_0=1$, $v_0=100\text{m/s}$ for alpha=3, beta=4, delta=2')
plt.grid()
plt.show()

```



From the plot, we see a decaying oscillation where the velocity reaches 0 at $x = 0$ when time goes to infinity. The numerical solution agrees with the phase portrait where all constants are positive leading to spiralling trajectories towards $x = 0$ suggesting that $x=0$ is a stable equilibrium point.

Driven and damped case (20 points)

- 1h. (5 pt) Now consider the damped and driven case, $\gamma \neq 0$. What is the equation of motion in this case? Can you develop this equation of motion fully from a potential (like we did above)? If so, what is the potential? If not, why not?

The equation of motion is,

$$\ddot{x} = \gamma \cos(\omega t) - \alpha x - \beta x^3 - \delta \dot{x}$$

We cannot obtain the equation of motion from a potential. The force function is dependent on time thus when the curl of the force is non-zero since time > 0. The force cannot be conservative hence there is no potential.

- 1i. (5 pt) Is it possible to plot the phase space for the driven and damped case? Why or why not? What solutions are available to you to understand the behavior of the system in phase space? *Here we are not looking for you to solve the problem, but to conduct research into how people make sense of the behavior of driven and damped systems.*

The phase space can be plotted however because of the time dependence in the equation of motion, we need to extend the phase space to 3D with time as the third dimension to prevent intersections in the trajectories in 2D. You can plot the full numerical solution with time on the vertical axis. The solutions available to us are the cases where one of the parameters or a combinations of parameters are zero such as the solutions of the undamped, undriven oscillator or the damped, driven oscillator. With this, you can see how the trajectories spiral or move away from equilibrium once the constants are non-zero.

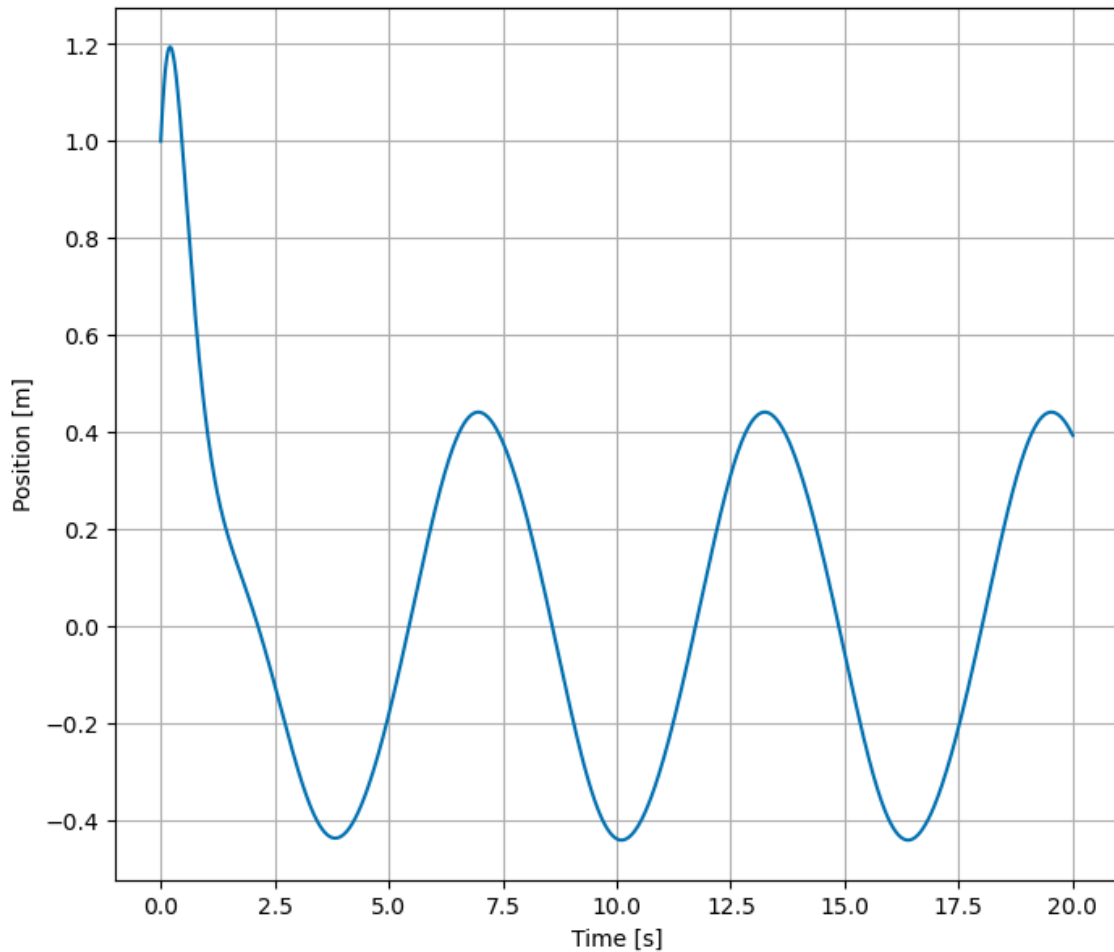
- 1j. (10 pt) **Here you are likely to need a numerical solver.** Solve the equation of motion for the driven and damped case for a reasonable choice of α , β , δ , γ , ω , and initial conditions. Plot the position as a function of time. What is the behavior of the system?

```
In [75]: alpha10 = 4
         beta10 = 3
         delta10 = 3
         gamma10 = 2
         omega10 = 1

         def DO_d_d(t, y, alpha10, beta10, delta10, gamma10, omega10): #d_d represents damp
             x, v = y
             return [v, -alpha10 * x - beta10 * x**3 - delta10 * v + gamma10 * np.cos(omega10 * t)]

         y0 = [1, 2] #Initial conditions are chosen to be x_0=1m, v_0=2m/s
         t_span2 = np.linspace(0, 20, 10000)
         sol_d_d = solve_ivp(DO_d_d, (0,20), y0, args=(alpha10, beta10, delta10, gamma10, omega10))
         x3 = sol_d_d.y[0]
         v3 = sol_d_d.y[1]
         plt.figure(figsize=(8,7))
         plt.plot(sol_d_d.t, x3)
         plt.xlabel('Time [s]')
         plt.ylabel('Position [m]')
         plt.title('Trajectory with $x_0$=1, $v_0$=100m/s for alpha=4, beta=3, delta=3, gamma=2, omega=1')
         plt.grid()
         plt.show()
```

Trajectory with $x_0=1$, $v_0=100\text{m/s}$ for $\alpha=4$, $\beta=3$, $\delta=3$, $\gamma=2$, $\omega=1$

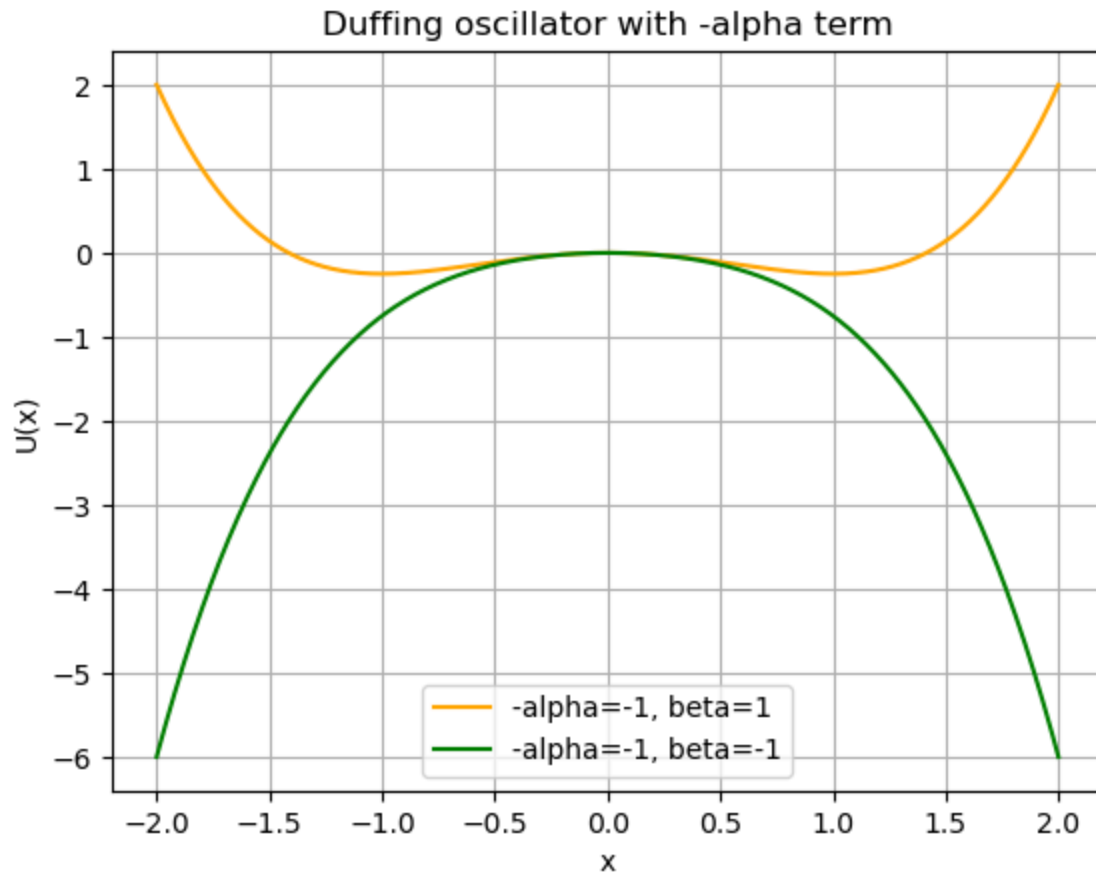


The transient oscillation quickly decays to zero due to damping. However, the driving force dominates and the trajectory becomes steady state.

Parameter Change (5 points)

- 1k. (5 pts) The Duffing oscillator can be either a soft or hard spring by changing the sign of the α term. What happens to the behavior of the system when you change the sign of α ? Can you explain this behavior in terms of the potential?

```
In [76]: alpha1 = 1
U4 = - m * alpha1 * x**2/2 + m * beta1 * x**4/4
U5 = - m * alpha1 * x**2/2 + m * beta2 * x**4/4
plt.plot(x, U4, color='orange', label='-alpha=-1, beta=1')
plt.plot(x, U5, color='green', label='-alpha=-1, beta=-1')
plt.xlabel('x')
plt.ylabel('U(x)')
plt.title('Duffing oscillator with -alpha term')
plt.legend()
plt.grid()
plt.show()
```

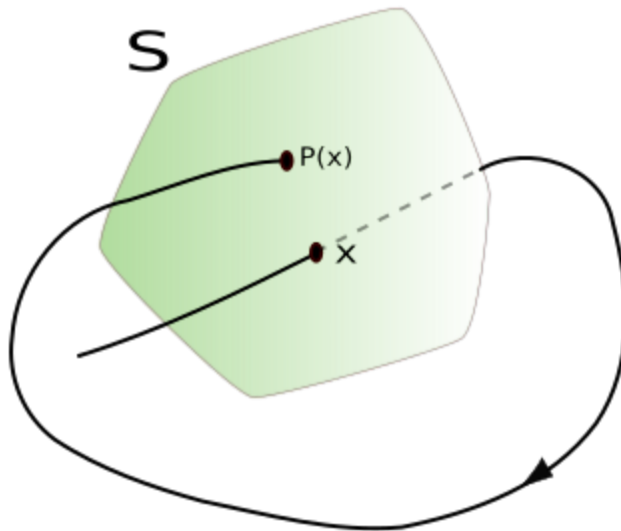


Changing the sign of a positive α to negative (β assumed to be positive) leads to the potential to have an unstable equilibrium point at $x=0$ and two stable equilibrium points. This is true from the conditions of stable and unstable equilibrium points.

This leads particles to move away from $x=0$ if they do not have sufficient energy to cross the barrier and oscillate around the either two stable equilibrium points.

If changing the sign of α leads to $-\alpha$ being the same sign as β , then the only equilibrium point of the potential is $x=0$. It is stable if $-\alpha$ is positive and unstable if $-\alpha$ is negative. Then a particle can oscillate around $x=0$ or move away from $x=0$ respectively.

Poincaré section (20 pts)



- 1l. (20 pts) A critical tool for understanding the behavior of the Duffing oscillator is the [Poincaré section](#).
 - **Implement a Poincaré section for the driven and damped case.** For choices of different parameters that **demonstrate qualitatively different behaviors**.
 - What can you claim about the behavior of the system given the Poincaré sections you created?
 - How does a Poincaré section help you understand the behavior of the system? *Here we are not looking for you to solve the problem, but to conduct research into how people make sense of the behavior of systems using Poincaré sections.*

Below is an example of the damped, driven 1D duffing oscillator. Use this as a template to do a few other types. As a part of this, you can generate different trajectories and do the following for them.

Poincare section for $\alpha = -1$, $\beta = 1$, $\gamma = 0.3$, $\omega = 1.2$

```
In [77]: #x = sol.y[0]
#v = sol.y[1]
#t = sol.t
def makeCrosspointsArray(x,v,c,transpose=False):
    crosspoints = []
    #c is the x value we slice for the poincare section.

    for i in range(len(x) - 1):
        if ((x[i]-c) * (x[i+1]-c)) <= 0:
            # (x[i+1]-c) crosses x=c then (x[i+1]-c) is negative so the product is
            x_avg = (x[i]+x[i+1])/2
            v_avg = (v[i]+v[i+1])/2
            #avg our values to get a middle estimate instead of a left estimate for
            if transpose == False: #allow transposing for better viewing.
                crosspoints.append([x_avg, v_avg])
            else:
```

```

        crosspoints.append([v_avg,x_avg])

crosspoints = np.array(crosspoints)
return crosspoints

```

```

In [78]: def trajAndPoincarePlot(alpha=-1,beta=1,delta=0.2,gamma=0.3,omega=1.2,y0=[1.0,0],t_
        """`c` is crossing line in x, and `t_interval` is a tuple from initial to final
        `y0` is a list containing position and velocity, respectively.
        Doesn't return anything, just adds the plot to the cache. After calling with ch
        t_span = np.linspace(*t_interval, 10000)
        sol_new = solve_ivp(D0_d_d, t_interval, y0, args=(alpha, beta, delta, gamma, om
        x = sol_new.y[0]
        v = sol_new.y[1]
        #do integration

        crosspoints = makeCrosspointsArray(x,v,c,transpose=transpose)
        #call function we just made, and use an x value that gets hit at multiple point
        fig, ax = plt.subplots(1,2, figsize=(12,8))
        ax[0].plot(sol_new.t,x)
        ax[0].set_xlabel(r'$t$ (s)')
        ax[0].set_ylabel(r'$x$ (m)')
        ax[0].set_title('Trajectory of Oscillation; Position-Time')
        ax[0].grid()
        #Plot trajectory

        ax[1].scatter(crosspoints[:,0], crosspoints[:,1]) #proper indexing to get arra
        if transpose == False:
            ax[1].set_xlabel(r'Position, $x$')
            ax[1].set_ylabel(r'VeLOCITY, $v$')
            if (c-0.02 <= min(crosspoints[:,0])) or (c+0.02 >= max(crosspoints[:,0])):
                ax[1].set_xlim((c-0.02,c+0.02))
        else: #pay attention to the actual labels being set here.
            ax[1].set_ylabel(r'Position, $x$')
            ax[1].set_xlabel(r'VeLOCITY, $v$')
            if (c-0.02 <= min(crosspoints[:,1])) or (c+0.02 >= max(crosspoints[:,1])): #
                ax[1].set_ylim((c-0.02,c+0.02))
        ax[1].set_title(f'Poincaré section at x = {c} (crossings)')
        ax[1].grid()
        #plot of poincare section.
        return

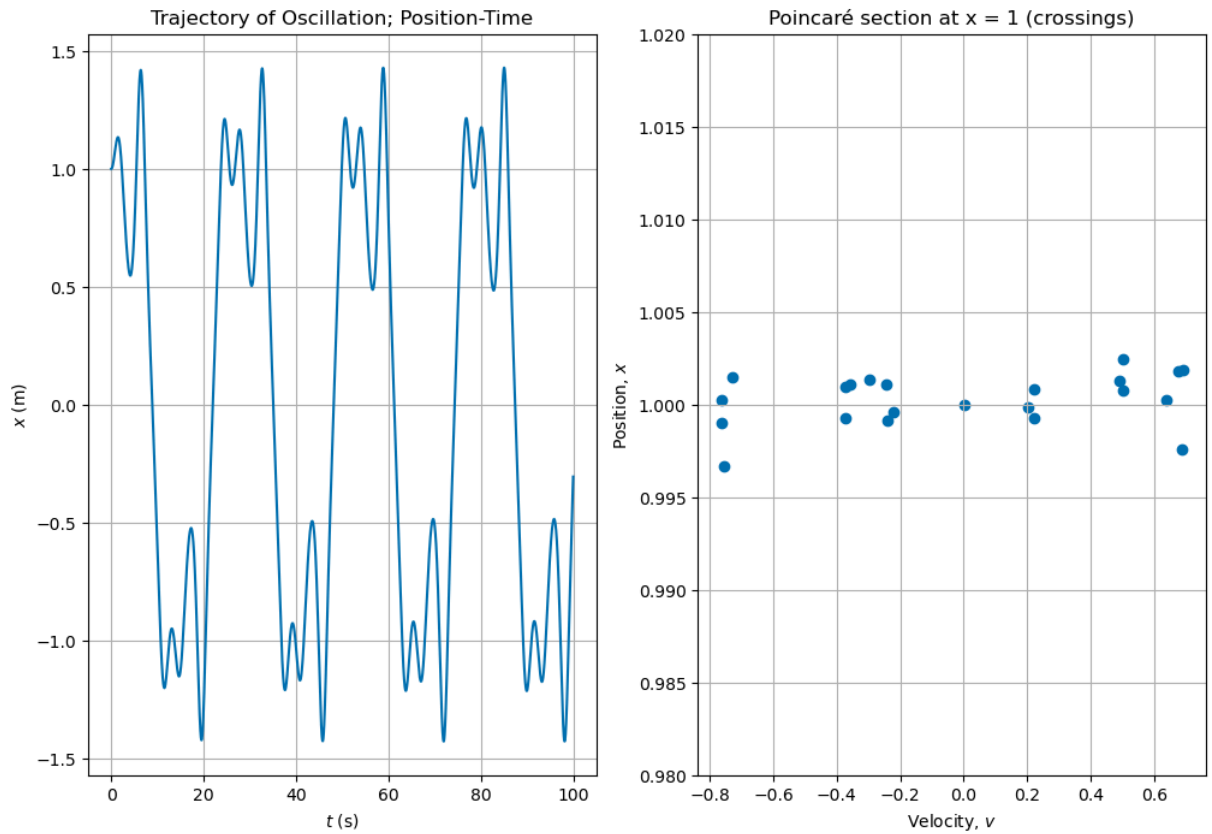
#function to call for easy access. Just plug in numbers and get graphs out.

```

```

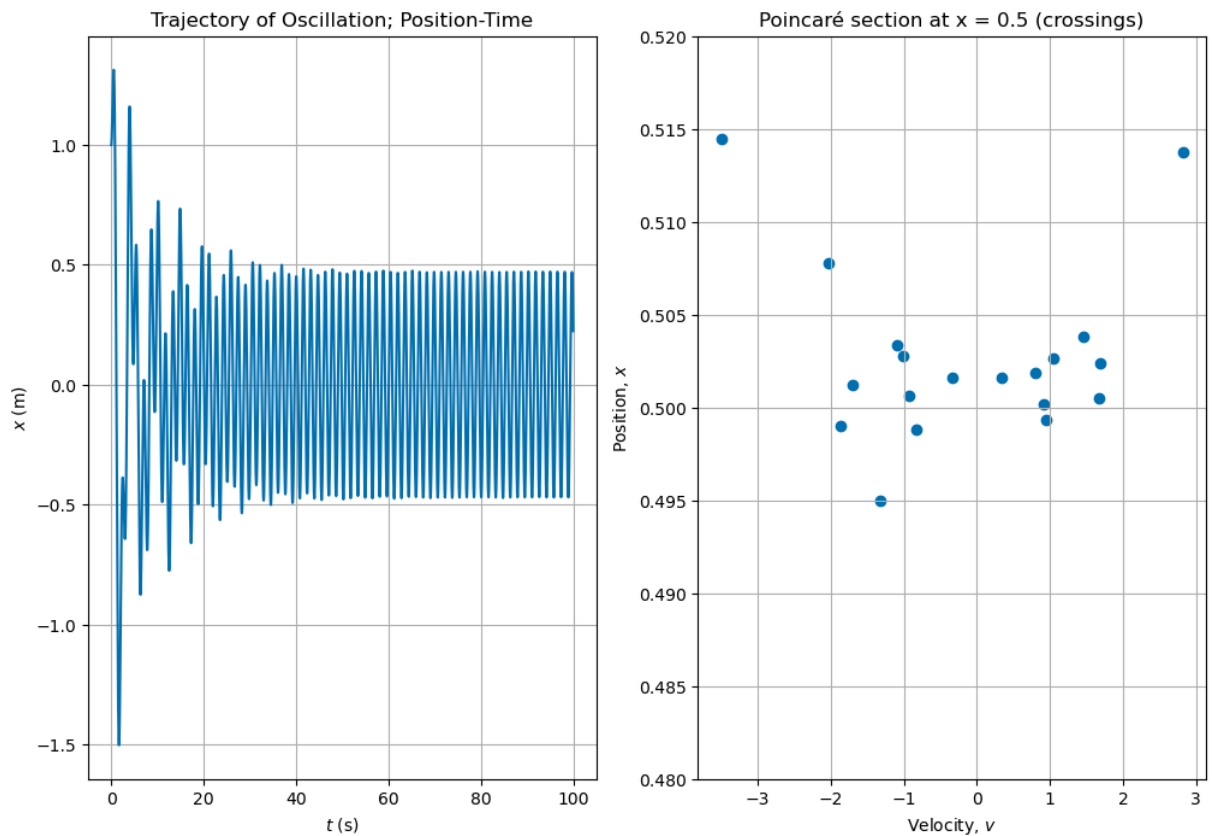
In [79]: #this is how you call the function to plot it.
        trajAndPoincarePlot(transpose=True)
        plt.show()

```



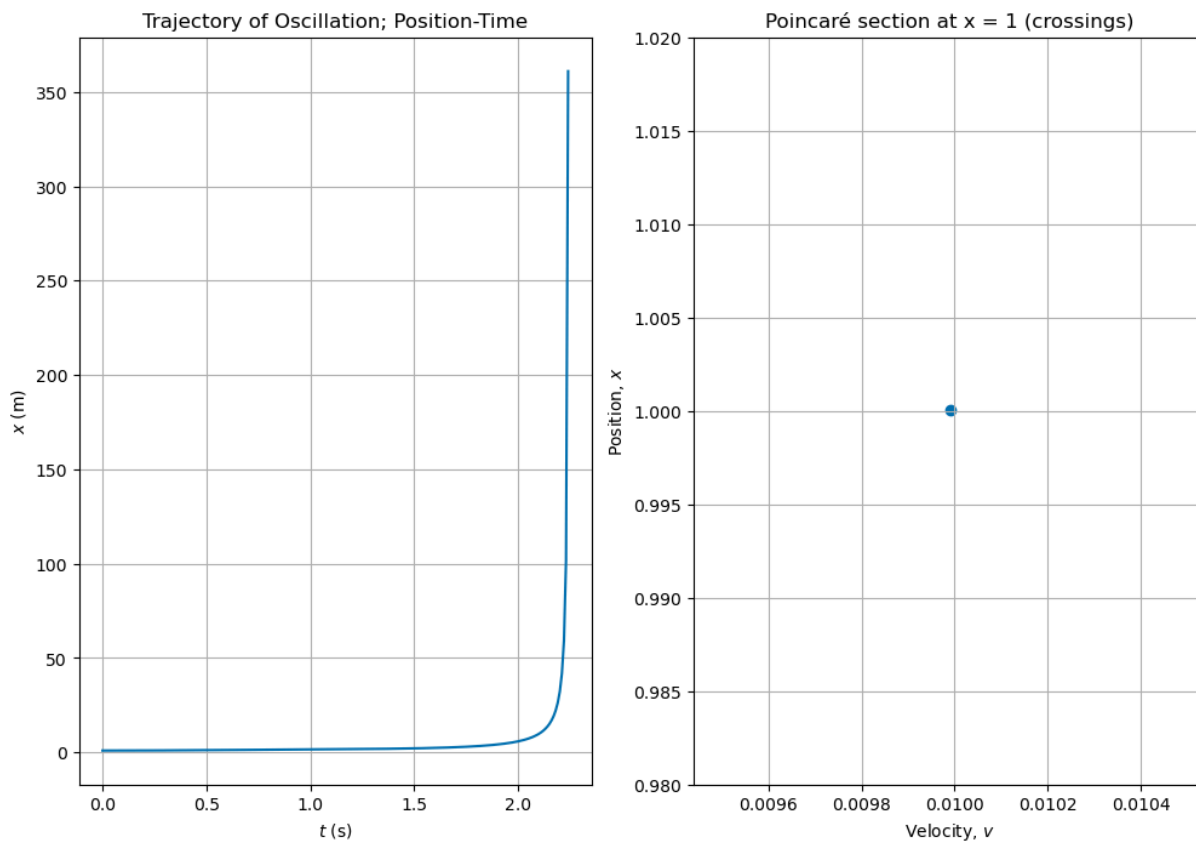
Poincare section for $\alpha = 1$, $\beta = 1$, $\delta = 0.2$, $\gamma = 7$, $\omega = 4$

```
In [80]: trajAndPoincarePlot(alpha=1,beta=1,delta=0.2,gamma=7,omega=4,y0=[1.0,0],t_interval=
plt.show()
```



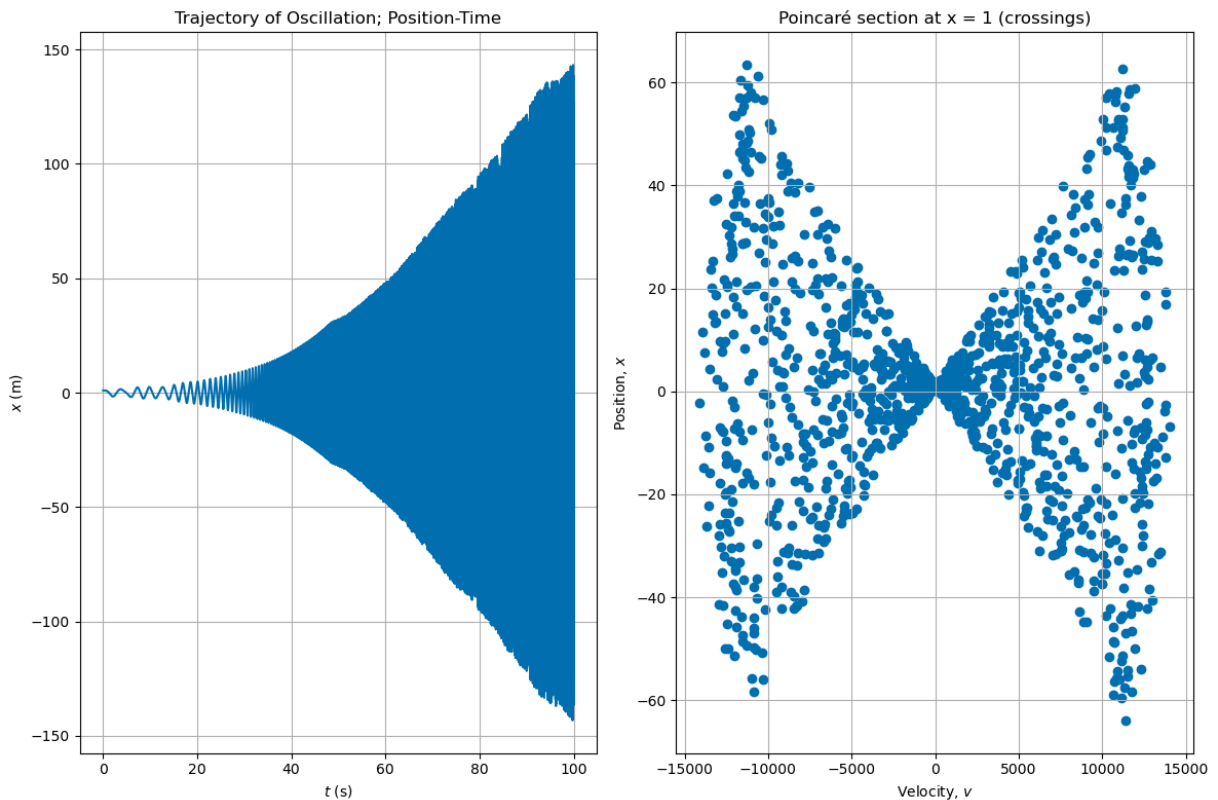
Poincare section for $\alpha = 1$, $\beta = -1$, $\gamma = 2$, $\delta = 0.2$, $\omega = 3$

```
In [81]: trajAndPoincarePlot(alpha=1,beta=-1,delta=0.2,gamma=2,omega=3,y0=[1.0,0],t_interval
plt.show()
```



Poincare section for $\alpha = 1$, $\beta = 1$, $\gamma = 2$, $\delta = -0.2$, $\omega = 3$

```
In [82]: trajAndPoincarePlot(alpha=1,beta=1,delta=-0.2,gamma=2,omega=3,y0=[1.0,0],t_interval
plt.tight_layout()
plt.show()
```



The behaviour of the systems are oscillatory trajectories that return to the same point in the cycle for all positive values of constants (even for negative α). We observe only one crossing at $x=1$ for negative β . Many oscillations for negative δ causes many crossing at $x=1$.

The number of trajectory intersections with the line $x=1$ varies with the constants of the EOM. The poincare sections are useful in giving a snapshot of the system at specific times and how the system may cross a certain point. It shows whether the system is periodic, non-periodic or quasi-periodic.